



A PROJECT REPORT ON

# JAP COUNTER APP

*A Mindful Mantra Chanting & Spiritual Habit-Tracking  
Android Application*

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MARWADI UNIVERSITY, RAJKOT

Information and Communication Technology  
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**CERTIFICATE**

This is to certify that the project entitled **JAP COUNTER APP** has been carried out by **Nikhil Bhanderi(92301733054)** under my guidance in partial fulfillment of the degree of Bachelor Engineering in Information and Communication Technology (7th Semester) of Marwadi University, Rajkot during the academic year 2026.

Date : 09-09-2026

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# 1. Introduction

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Jap Counter App is a serene, spiritually themed Android application built to support mindful chanting, mantra counting, and spiritual habit tracking. In many Indian spiritual traditions, practitioners repeat a mantra or name of the divine a fixed number of times (commonly 108 or 1008) using a physical rosary (mala). This application digitises that practice, giving users a calm, distraction-free way to count their chants while preserving the meditative spirit of the exercise.

The app was designed with a warm Saffron, Sandalwood, and Cream colour palette to keep the experience visually soothing, and is built using modern Android development practices — Jetpack Compose for the UI, an MVVM architecture, and local-first data persistence — so that it remains fast, private, and fully usable without an internet connection.

## Objectives

- Provide a simple, elegant, and distraction-free digital counter for mantra chanting (jap).
- Allow users to set custom mantra names and target counts (108, 1008, or any custom value).
- Maintain a complete history of chanting sessions, including in-progress and completed sessions.
- Allow users to resume an interrupted session at any time without losing progress.
- Celebrate the completion of a chanting goal with a dedicated, encouraging completion screen.
- Deliver a calm, spiritually appropriate visual design suitable for meditation and prayer.

## 2. Features

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### Serene Animated Splash Screen

The app opens with a glowing “Om” emblem accompanied by a gentle breathing pulse animation and a soothing gradient background, setting a calm and spiritual tone before the user even begins chanting.

### Mindful Jap Counter

- An intuitive, large circular tap counter that responds to every touch with a satisfying, calming interaction.
- A real-time percentage progress ring that fills as the user gets closer to their target.
- Quick target-count shortcuts for the traditional counts of 108 and 1008, plus support for any custom target.
- A dropdown with auto-suggestions of previously used mantra names, so recurring practices are quick to start.

### History & Task Tracking

- A complete, chronological history of chanting sessions — both completed and in-progress.
- Sessions are grouped by date (Today, Yesterday, and so on) with completion timestamps and progress indicators.
- Any in-progress session can be resumed at any time, exactly where the user left off.
- Individual history records can be deleted, or the entire history cleared, with a confirmation dialog to prevent accidental data loss.

### Completion Celebration

When a chanting target is reached, the app transitions to a dedicated completion screen that celebrates the achievement, with one-tap options to restart a new session or view the session in the history log.

### Spiritual Aesthetics

Every screen uses a warm Saffron, Sandalwood, and Cream colour palette, chosen specifically to promote tranquility and focus, making the app feel at home alongside meditation, prayer, and other mindfulness practices.

### 3. Technology Stack & Architecture

Jap Counter App is built entirely with modern, native Android tooling. The technology choices prioritise a smooth, reactive UI, reliable offline-first data storage, and a clean separation of concerns between the interface and the underlying business logic.

| Category              | Detail                                                                                                                              |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Language              | Kotlin                                                                                                                              |
| UI Toolkit            | Jetpack Compose (Material 3)                                                                                                        |
| Architecture          | MVVM (Model-View-ViewModel) + Repository Pattern                                                                                    |
| Local Persistence     | Room Database (chanting history & session logs) and DataStore Preferences (active session state, current counts, user profile data) |
| Asynchronous Handling | Kotlin Coroutines & Flow / StateFlow                                                                                                |
| Navigation            | Jetpack Compose Navigation (NavHost)                                                                                                |
| Minimum SDK           | Android 7.0 (API Level 24)                                                                                                          |
| Target SDK            | Android 15 (API Level 35)                                                                                                           |

#### Architecture Overview

The application follows the MVVM pattern layered with a Repository. The UI (Compose screens) observes StateFlow exposed by a central JapViewModel, which owns all business logic and mediates between the UI and the data layer. The Repository abstracts the Room database and DataStore preferences behind a single API, so the ViewModel never talks to persistence directly. This keeps the UI layer simple and reactive, and makes the data layer easy to test or swap out.

## 4. Project Structure

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The codebase is organised by responsibility, following standard Android/Compose conventions, which keeps data, UI, and navigation cleanly separated:

### **data/**

JapPreferences.kt — DataStore preferences for the active session

JapSession.kt — Session data model

### **local/**

JapDatabase.kt — Room database definition

JapHistoryDao.kt — Room DAO operations

JapHistoryEntity.kt — History table entity

### **repository/**

JapHistoryRepository.kt — Repository layer for data operations

### **ui/**

components/NavComponents.kt — Custom app bars & navigation headers

history/JapHistoryScreen.kt — Jap history & past records screen

navigation/JapNavGraph.kt — Compose navigation graph & routes

### **screens/**

SplashScreen.kt — Animated spiritual splash screen

WelcomeScreen.kt — User onboarding screen

SetupScreen.kt — Mantra & target configuration

CounterScreen.kt — Main interactive jap counter screen

CompletionScreen.kt — Jap completion celebration screen

### **theme/**

Color.kt, Theme.kt, Type.kt — Saffron & Sandalwood palette, Material 3 theme, typography

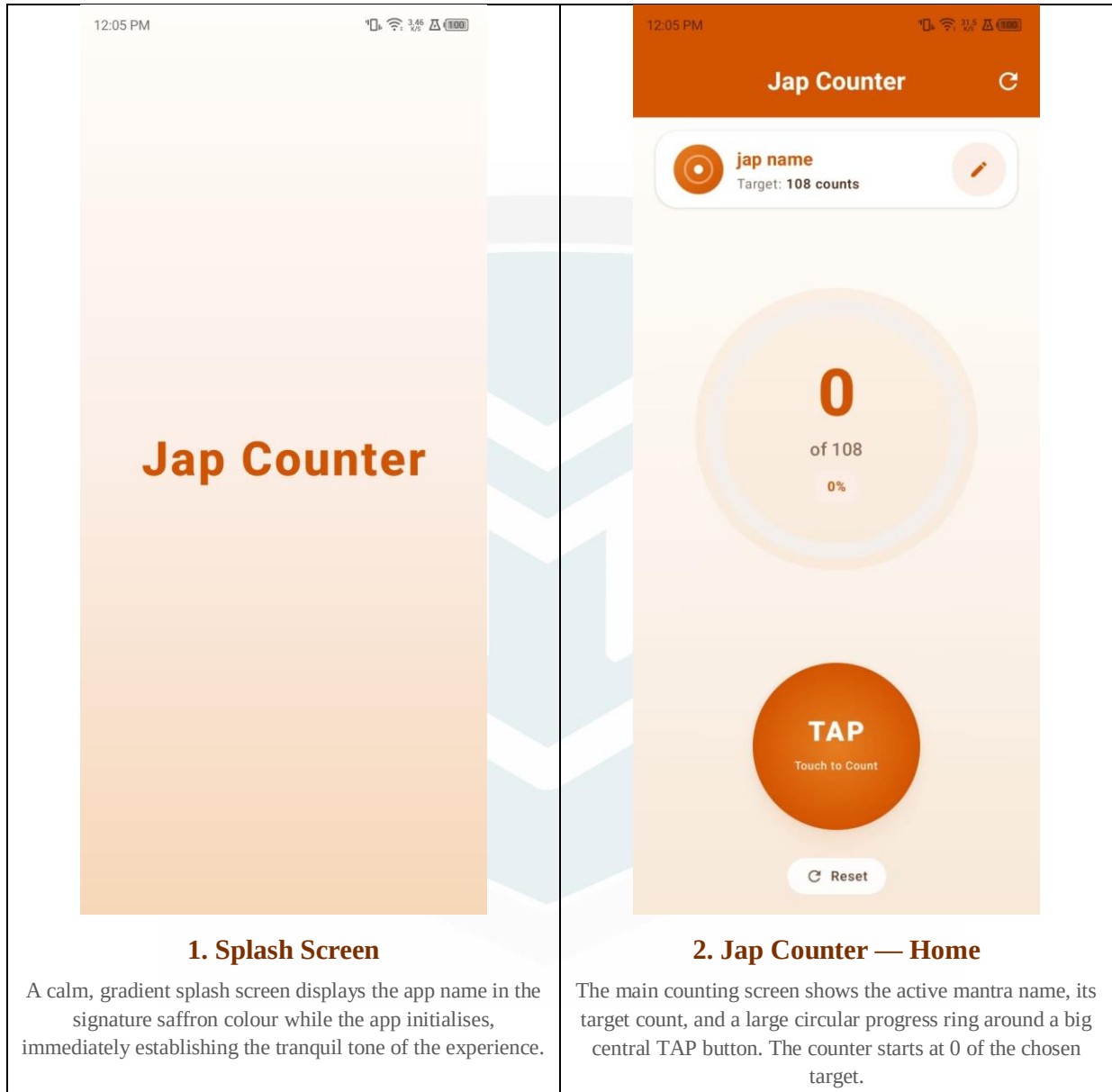
viewmodel/JapViewModel.kt — Central ViewModel handling business logic

### **MainActivity.kt**

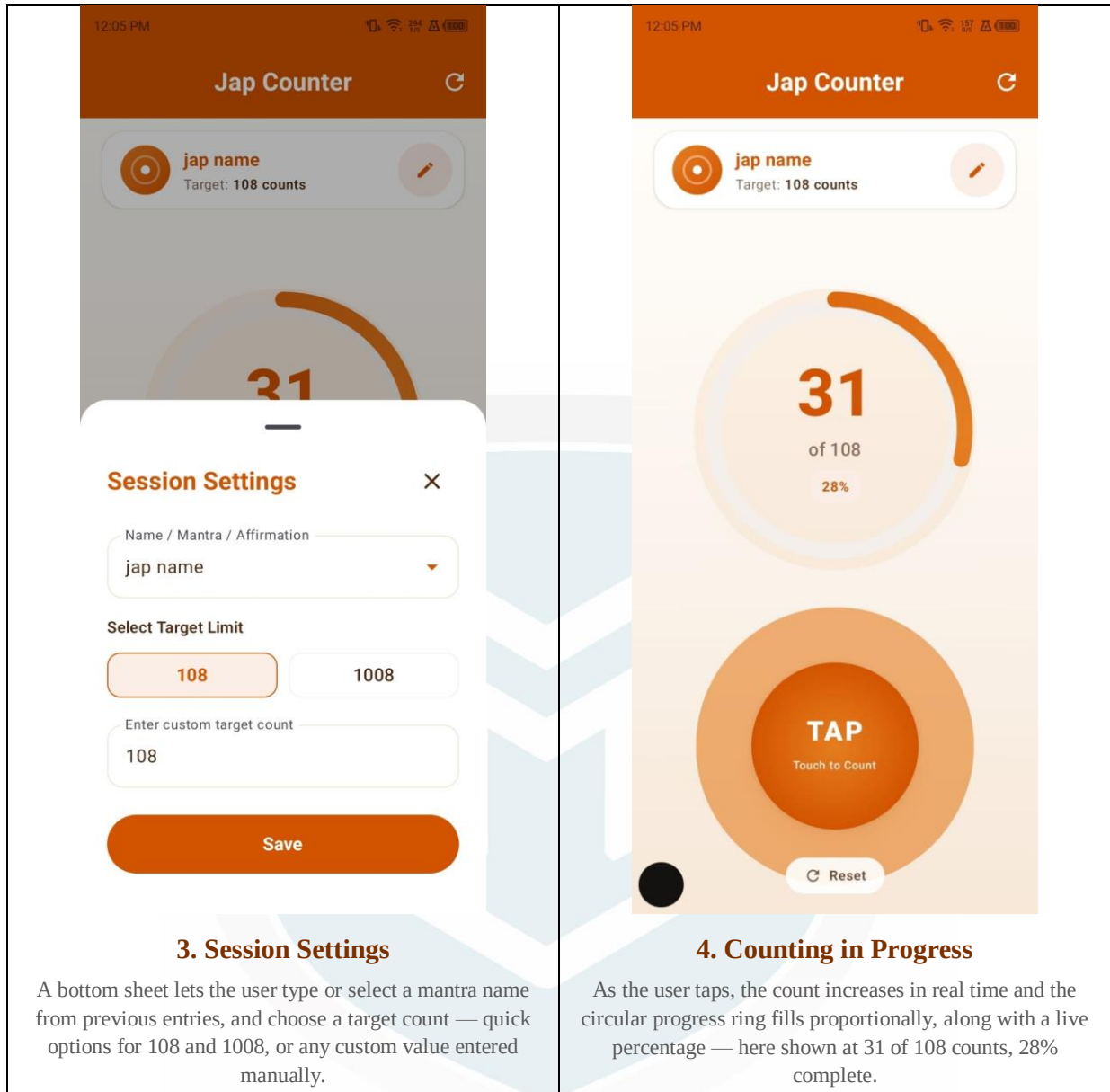
## 5. Screenshots & Description

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The following screens illustrate the core user journey of Jap Counter App — from the moment it launches, through configuring and running a chanting session, to reviewing past sessions in history.





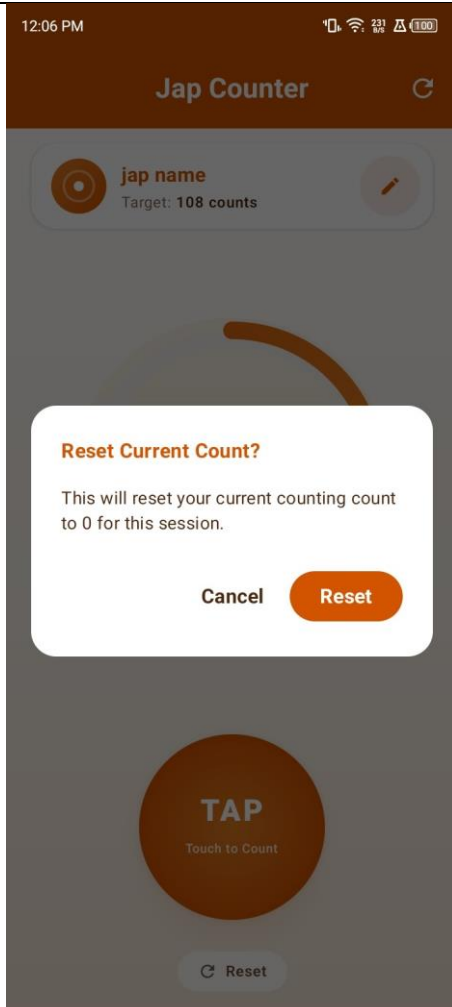


### 3. Session Settings

A bottom sheet lets the user type or select a mantra name from previous entries, and choose a target count — quick options for 108 and 1008, or any custom value entered manually.

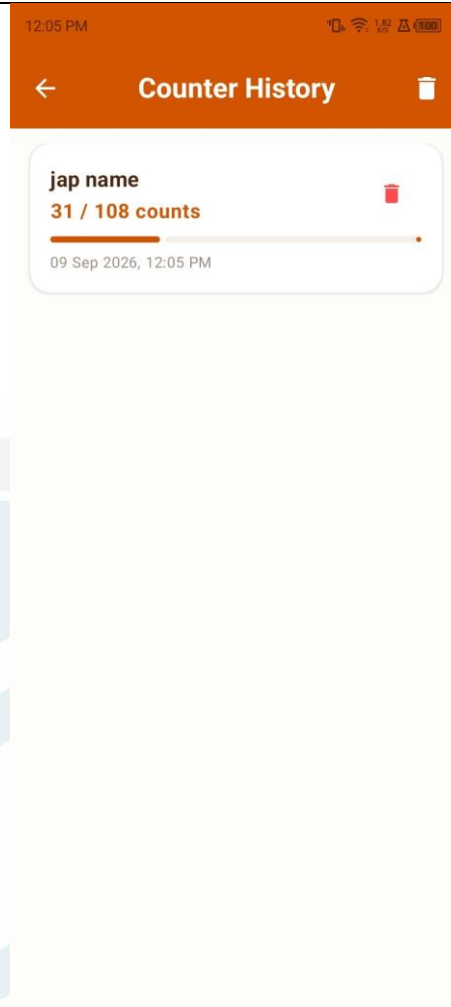
### 4. Counting in Progress

As the user taps, the count increases in real time and the circular progress ring fills proportionally, along with a live percentage — here shown at 31 of 108 counts, 28% complete.



### 5. Reset Confirmation

Tapping Reset opens a confirmation dialog before the current session's count is cleared back to zero, preventing accidental loss of progress.



### 6. Counter History

The history screen lists past chanting sessions grouped with date and time stamps, showing each mantra name, its progress (e.g. 31 / 108 counts), and a visual progress bar; entries can be deleted individually or cleared entirely.

## 6. Future Scope

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Jap Counter App already covers the essentials of digital mantra counting. A number of enhancements could extend it into a fuller spiritual companion app:

- **Cloud Backup & Sync** — Optional account-based sync so chanting history and streaks carry over across devices.
- **Daily Reminders & Streaks** — Configurable notifications and a streak tracker to encourage a consistent daily practice.
- **Audio-Guided Chanting** — Built-in audio tracks or a metronome-style beat to pace group or guided chanting sessions.
- **Voice / Mic-Based Counting** — Automatic count detection through spoken repetition, for hands-free jap during other activities.
- **Wear OS / Smartwatch Support** — A companion watch app or tile so counting doesn't require holding a phone.
- **Widgets** — A home-screen widget for one-tap counting and at-a-glance progress without opening the app.
- **Multi-language Support** — Localisation of the interface and mantra library into Hindi, Gujarati, and other regional languages.
- **Meditation Timer Integration** — An optional silent meditation/breathing timer that can precede or follow a chanting session.
- **Data Export & Analytics** — Exportable history (CSV/PDF) and simple analytics such as weekly/monthly chanting trends.
- **Dark Mode** — A low-light, dark-themed variant of the palette for night-time or early-morning practice.

## 7. Conclusion

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Jap Counter App demonstrates how a simple, everyday spiritual practice can be thoughtfully translated into a digital experience without losing its calm and meditative character. Built on a modern MVVM architecture with Jetpack Compose, Room, and DataStore, the app is fast, works fully offline, and keeps the user's chanting history safely on-device. Its warm Saffron and Sandalwood aesthetic, intuitive tap-to-count interaction, and thoughtful history tracking make it a genuinely useful companion for anyone maintaining a regular jap or mantra practice, while its clean, modular codebase leaves plenty of room for the future enhancements outlined above.

